Finding product launches in SEC filings

CAP5619 - Artificial Intelligence and Finance | Assignment 2
Methodology and results | November 2, 2026

SOURCE-CHECKED PRODUCTS	LLM-ANALYZED FILINGS	NEW API ATTEMPTS
100	187	153 / 180

Research question and scope

This project identifies original announcements of completed product or distinct new service launches in SEC Form 8-K filings. The output has exactly company_name, stock_name, filing_time, new_product, and product_description; descriptions contain fewer than 180 characters. The written Assignment 2 handout and course examples guide the Python, SEC collection, LLM extraction, and reporting approach. The discovery pool spans 2020-02-05 through 2026-10-30. The sample is purposive rather than representative.

Data collection and model specifications

The merged discovery pool contains 1417 unique filing URLs. SEC full-text searches used launch and availability phrases, including announced the launch, today launched, introduces/new products, and launches/now available. Pagination limits and retrieved document-hit counts are logged; document hits are deduplicated to filing URLs. The main 8-K and readable EX-99 exhibits are fetched with an identified User-Agent, sequential pacing, local caching, and limited transient-error retries. The request ledger records 153 new model attempts. The configured model is gpt-6-luna through OpenAI's Responses API (recorded identifiers: gpt-6-luna). Each call uses strict JSON Schema, at most 60,000 source-text characters and 10,000 output tokens. No fine-tuning or explicit temperature setting is used. Model weights and training details were not independently available. Actual token usage retained for these attempts totals 617,767 input and 56,460 output tokens; the report does not infer a dollar charge.

Extraction and review method

Free screening ranks launch language near exhibit openings and penalizes earnings, financing, rebrand, website, and other non-product headlines. Selected filings are analyzed one per model request. Named Entity Recognition (NER) is performed by the LLM: it identifies company names as ORGANIZATION entities and newly announced products as PRODUCT entities, returning company_name and new_product within structured fields. These are semantic entity roles, not separately emitted BIO tags. No separate NER model was trained; issuer names are subsequently normalized to SEC metadata. The prompt excludes historical mentions, planned developments, prototypes, acquisitions, issuer financing, and channel expansions. It requests the source URL, a contiguous launch quote, announcement date, confidence, and rationale alongside the five data fields. Python supplies SEC's acceptance timestamp, validates the ticker and field types, rejects long descriptions, and checks the quote against supplied text. Names are deduplicated within a ticker using normalization and close-spelling comparison. An AI assistant groups strengths, configurations and suite variants rather than expanding them into extra rows, reads the original exhibit, checks the product description and current-launch context, and records an explicit source-review decision. Only source-checked or human-verified rows can enter the final CSV. The source_checked label explicitly denotes AI-assisted source review, not independent human verification.

Quality, results, and limitations

Observed problems and corrections

Search terms also retrieved past launches in earnings summaries and unrelated corporate announcements. High model confidence was insufficient: later paragraphs sometimes revealed beta status or future commercial availability despite a launch headline. Source review excluded these and historical recaps. Six earlier uncertain cases were reassessed; four were rejected and two remain excluded. Genius retains VRSSD: SEC exhibits document its March 2025 adoption and April 17 use, but the April 30 launch release lists VRSSF. The effective symbol on May 2 is unresolved; no current-ticker inference was used. Details and source references are in audit/genius_ticker_review.json. SEC inline HTML added spaces around punctuation; quote comparison normalizes this without changing words. Some model quotes omitted intervening text and some descriptions were too long. Documented source-based corrections preserve raw responses and require another offline check. An attached launch release may report initial shipments before SEC acceptance; historical recaps are excluded.

Results and completion status

The project retains 135 candidate product rows, including the unchanged 11-row legacy baseline. The final export contains 100 source-checked product rows from 62 companies and 85 distinct issuer/acceptance-time pairs. Across preserved history and this collection, 187 unique filings from 133 SEC CIKs (135 ticker labels) have successful LLM analysis logs. 1068 filings have successful free screening records; 1100 distinct filings were screened or LLM-analyzed in total. These categories overlap and must not be added. Historical logs include filings outside the current discovery pool. Review outcomes are 100 source_checked, 9 uncertain, 26 rejected, and 0 pending. The 100-entry objective is met, and the 100-LLM-filing objective is met. The product shortfall is 0 rows. No rows were invented or duplicated to satisfy either target.

Reproducibility, limitations, and AI assistance

Course methods include SEC text collection, contextual prompting, entity extraction, and structured output. Concrete budget limits, journals, locks, hashes, and preservation checks are additional engineering decisions, not techniques specifically required by the professor. This is LLM inference, not model training; the written handout does not require Weights & Biases, and none is integrated. Complete raw outputs, recovery journals, and development history remain in a private archive. This public export retains checkpoints, SEC source packets, decisions, and hashes for offline validation. The original workflow used atomic journals for recovery. Its ledger reserves requests before sending, disables SDK retries, and enforces the user's cumulative 180-attempt ceiling across restarts; failures consume a slot. This batch used 33 attempts. The initial 60-attempt grant was increased twice by explicit 60-attempt approvals. Paid extraction stops once 100 source-supported products pass review. All 73 previously source-checked rows and prior audit history were preserved. Windows Credential Manager holds the API key. The README documents execution and offline tests. The attached CSV and report state actual progress and remaining requirements. Screening may miss launches in financial releases or beyond the first exhibits, and truncated inputs need review. No labeled benchmark was available, so precision, recall, and exhaustive originality are not claimed. Announcements are issuer statements, not independent verification of efficacy or performance. Codex assisted code development, tests, extraction design, source review, and report preparation. The student should review the source, decisions, and instructor AI-use rules before submission.

GitHub repository: https://github.com/zackjarmstrong/CAP5619-Assignment2-SEC-Product-Launches. Sources: Assignment 2 handout and course examples reviewed privately (not redistributed); SEC EDGAR filing/exhibit URLs are recorded per product in product_evidence.csv. API documentation: developers.openai.com/api/reference/python; developers.openai.com/api/docs/guides/structured-outputs; developers.openai.com/api/docs/models/gpt-6-luna. Accessed November 1, 2026.